

‘Does the presence of corruption inhibit a nation's growth prospects?’

**An investigatory comparative economic literature between post colonization
Africa vs Asia**

Sandeep Rai
Columbia University
Master of Arts
Thesis

Abstract

This paper tests whether institutional quality, specifically corruption, accounts for a meaningful share of the Africa-Asia growth gap. Using panel data on fifteen countries from 1998 to 2023, we estimate the cross-country relationship between perceived corruption and the country risk premium, decompose the Africa-Asia growth gap into capital, labour, and total factor productivity components, examine six anti-corruption reform events for spread responses, and construct a new dataset of reform follow-through across twenty-two events spanning four continents.

Four results stand out. First, in the cross-country dimension, a one-unit deterioration on the Worldwide Governance Indicators control-of-corruption score is associated with a 53% higher log country risk premium (coefficient 0.53, $p = 0.014$, $N = 246$, country-clustered standard errors), corresponding to approximately 180 basis points of the 894 basis point Africa-Asia spread gap, or twenty percent of the gap. Within-country, within-year variation in corruption does not move the spread proxy (TWFE coefficient 0.12, $p = 0.58$), consistent with institutional quality acting as a level effect on country risk rather than a short-run driver.

Second, total factor productivity carries 55.6% of the Africa-Asia growth gap, and a panel regression of log TFP on the inverted WGI score with country and year fixed effects and macro controls returns a coefficient of -0.137 ($p = 0.010$, $N = 250$). A one-unit WGI deterioration corresponds to a 12.8% lower TFP. The TFP-corruption channel is the strongest result in the paper.

Third, the six anti-corruption reform events display the heterogeneity the institutional-architecture argument predicts. Indonesia's KPK (2003) is associated with a 19.5% drop in the lending-rate proxy ($p < 0.001$) and South Korea's consolidation window (2003 proxy) with a 16.0% drop ($p = 0.005$). Nigeria's EFCC, instituted in the same year as Indonesia's KPK and with the same nominal form, produces no measurable response ($p =$

0.56). Two events (Brazil Lava Jato 2014, Rwanda 1998-2008) show post-treatment increases that we attribute to macro confounding rather than to the reforms themselves.

Fourth, we introduce a new institutional-quality variable, a composite follow-through index for anti-corruption reform, and code it across twenty-two reform events spanning four continents (1991-2019). Each one-point increase on the index is associated with a 3.5 percent larger compression in the post-reform lending rate ($p = 0.035$, $N = 18$). The variable operationalises the institutional-architecture argument that has been load-bearing in the qualitative case-study literature for decades. The corruption tax is real and measurable. It is not the dominant driver of the Africa-Asia growth gap, but it accounts for a meaningful share of the cross-country spread differential and the TFP gap, and its measured effect varies systematically with the political conditions that determine whether anti-corruption reforms are allowed to function.

1. Introduction

This paper asks whether corruption inhibits a nation's growth prospects. The question has been the subject of empirical work for three decades, beginning with Mauro (1995) and extending through the institutional-quality literature that followed Acemoglu, Johnson, and Robinson (2001, 2002) and Rodrik, Subramanian, and Trebbi (2002). We approach it from two directions: by examining the cross-country relationship between perceived corruption and the country risk premium that sovereign bond markets price, and by decomposing the Africa-Asia growth gap into the factor inputs and the residual that the literature has long interpreted as total factor productivity.

Two empirical observations motivate the analysis. The first is that since the early 2000s, sovereign spreads and lending-rate proxies for emerging market debt have shown a stable cross-country relationship with corruption indices, holding macro fundamentals constant. Countries with lower Corruption Perceptions Index scores pay more to borrow. What has been underweighted in the literature since is the parallel signal that markets themselves have been producing. Sovereign yields contain a price for institutional quality, and the price has been telling us approximately the same thing the academic literature has been arguing.

The second observation is that within our sample, anti-corruption reform events that occurred at the same time produced sharply different lending-rate movements depending on the institutional architecture that received them. Indonesia and Nigeria both established anti-corruption commissions in 2003. By 2008 the Indonesian KPK had produced a series of high-profile prosecutions that contributed to a measurable tightening of the country's lending-rate proxy; Nigeria's EFCC, beset by political interference and limited prosecutorial independence, produced no comparable spread movement.

This paper takes both observations seriously. Working with a panel of fifteen countries from 1998 to 2023, we estimate the cross-country institutional risk premium, decompose the

Africa-Asia growth gap into capital, labour, and total factor productivity components, examine the six anti-corruption reform events for spread responses, and develop the institutional-architecture and political-feasibility analysis country by country.

1.1 The puzzle

In the years immediately following decolonisation, the trajectories of Africa and Asia ran in parallel. The countries that became known as the Asian Tigers began the 1960s with income per capita below many sub-Saharan African economies. By 2023 the divergence was substantial. South Korea's GDP per capita exceeded thirty thousand dollars in 2015 dollars; Ghana's remained below two thousand.

In our sample over the 1998 to 2019 window, the eight African economies averaged real GDP per worker growth of approximately 2.35 percent per year, while the five Asian economies averaged 3.08 percent per year. The 0.73 percentage-point per year gap is smaller than the popular discourse on African stagnation often implies, but it is sustained, and at compound rates it produces the income gap that motivates the question.

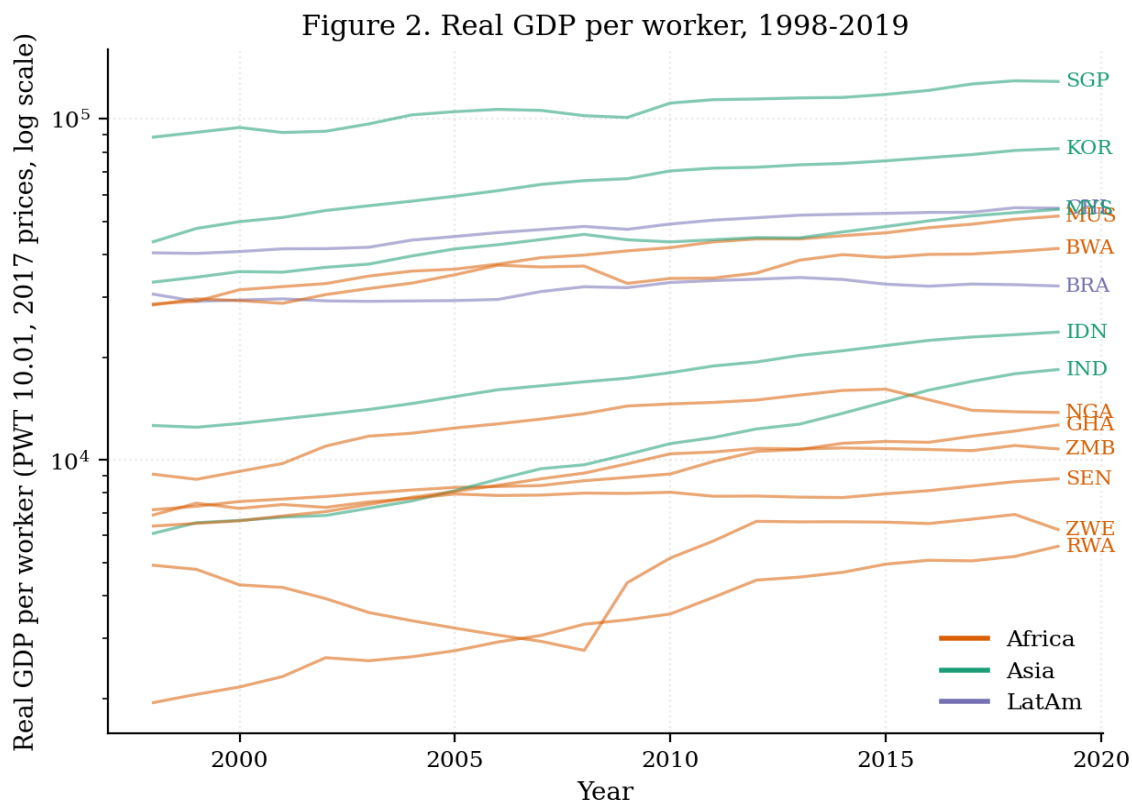


Figure 2. Real GDP per worker by country (1998-2019, PWT 10.01). African countries in orange, Asian countries in green, Latin American controls in purple. Log scale.

1.2 The thesis claim

Institutional quality, indexed by perceived corruption, is one of the central drivers of the Africa-Asia growth gap. It is not the only driver. The empirical share of the gap that loads on corruption in our specifications is meaningful but not dominant. The corruption channel operates principally as a cross-country level effect on the country risk premium and on total factor productivity, with within-country variation playing a smaller role. The gap is politically tractable. This paper investigates how.

1.3 The contribution

This paper makes five moves. First, we estimate the cross-country institutional risk premium. Holding macro fundamentals constant, a one-unit deterioration on the inverted WGI control-of-corruption score is associated with a 53 percent higher log country risk premium. Second, we decompose the Africa-Asia growth gap, attributing 55.6 percent to total factor productivity, which loads on corruption with a coefficient of -0.137. Third, we examine six anti-corruption reform events as quasi-experiments. Fourth, we introduce a new institutional-quality variable: a composite follow-through index for anti-corruption reform, coded across twenty-two reform events, with the variable predicting the post-reform lending-rate response at $p = 0.035$. Fifth, we analyse the political feasibility of institutional reform country by country.

1.4 The framing

We treat institutional quality as a risk premium that crowds out investment, analogous to factor crowding in financial markets. Sovereign bond markets price institutional quality cross-sectionally. Their pricing represents the market's collective estimate of corruption-related risk attached to a country's debt service. The framing is informative but the data limitations are real. We do not have proprietary EMBI sovereign-spread data for the full sample. We use the lending rate minus the US ten-year Treasury yield as a country-risk proxy, supplemented by IMF real long-term government bond yields where available.

1.5 The roadmap

Section 2 reviews the literature. Section 3 describes the data and identification strategy. Section 4 presents the panel regression results. Section 5 presents the growth-accounting decomposition. Section 6 introduces the follow-through dataset and reports the cross-event regression. Section 7 develops the institutional reform mechanism analysis. Section 8 analyses political feasibility country by country. Section 9 discusses limitations honestly. Section 10 concludes.

2. Literature review

The literature on corruption and growth has accumulated across four overlapping conversations: the investment-channel framing of corruption's effects on growth, the institutional-quality framing of the deeper drivers of cross-country income differences, the rival explanations centred on geography, ethnic fractionalisation, and resource endowment, and the Africa-specific empirical work that has tested cross-country findings against the continent's particular history. We organise the review around these four conversations and close with a fifth observation: the literature has underweighted the parallel signal that sovereign bond markets have been producing for three decades.

2.1 Corruption as a tax on investment

The investment-channel framing originates with Mauro (1995), who used ethnolinguistic fractionalisation as an instrument for corruption across approximately seventy countries and found that corruption depressed investment, with the investment channel accounting for the bulk of corruption's effect on growth. Bardhan (1997), Tanzi and Davoodi (1997), Mauro (2004), Treisman (2000), and Lambsdorff (2003) refined and extended the channel: corruption distorts the composition of public investment, persists in self-reinforcing equilibria, correlates with long democratic exposure and colonial origin, and affects capital inflows and domestic investment through separate channels.

2.2 Institutional quality as the underlying driver

A parallel literature argued that corruption was a symptom and the deeper variable was institutional quality. Acemoglu, Johnson, and Robinson (2001, 2002) used colonial settler mortality as an instrument and found large effects on long-run income per capita, with the reversal of fortune between historically rich and poor regions tracking the institutions colonisers installed rather than geography or culture. Rodrik, Subramanian, and Trebbi (2002) ran a horse race between institutions, geography, and trade and concluded institutions trumped the rivals. Kaufmann, Kraay, and Mastruzzi (2007) consolidated institutional measurement into the WGI; La Porta, Lopez-de-Silanes, and Shleifer (2008) reviewed the legal-origins literature.

2.3 Rival theories considered

Two rival explanations of cross-country growth differences deserve direct engagement. The first is geography. Sachs (2003) and Sachs and Warner (1995) argued that tropical disease burden, distance from coast, and natural-resource dependence explained a substantial share of the African growth shortfall. The framing has weaknesses: Botswana, Mauritius, and Singapore contradict the prediction. Rodrik, Subramanian, and Trebbi (2002) showed geography's apparent effect shrank substantially once institutional quality was instrumented. The second is ethnic fractionalisation. Easterly and Levine (1997) argued that ethnic diversity in sub-Saharan Africa produced low public goods provision. The argument has force where ethnic boundaries map onto patronage networks. We treat fractionalisation as a control rather than as a competing primary explanation.

2.4 Africa-specific corruption literature

Gyimah-Brempong (2002) replicated Mauro's findings within African panel data, ruling out the possibility that the global relationship is driven entirely by between-region variation. Humphreys and Bates (2005) added the political-economy conditionality that bears directly on Sections 7 and 8. Uslander (2008) developed the trust channel as a complement to the investment channel.

2.5 The underweighted angle

The literature has paid less attention to the parallel signal that sovereign bond markets have been producing for three decades. Cantor and Packer (1996) established that sovereign credit ratings priced governance quality. Hilscher and Nosbusch (2010) decomposed emerging-market spreads into global and country-specific components and found a substantial country-specific component that tracked governance variables. This paper bridges the gap by asking what spreads have priced for our fifteen countries between 1998 and 2023.

3. Data and methods

The empirical work rests on three pieces of infrastructure: a balanced panel of fifteen countries observed annually between 1998 and 2023, a set of country-risk-premium proxies built from World Bank Development Indicators and IMF DataMapper data, and a set of six reform-event dates that organise the difference-in-differences analysis.

3.1 Sample

The fifteen-country sample contains eight African economies (Nigeria, Senegal, Ghana, Zambia, Zimbabwe, Rwanda, Botswana, Mauritius), five Asian economies (Singapore, India, South Korea, Indonesia, Malaysia), and two Latin American controls (Brazil, Chile). The African block covers high-corruption rent-extracting cases, middle-band economies under reform pressure, the cases treated as exceptions in the broader development literature, and the post-conflict institutional rebuild case. The Latin American controls anchor the DiD analysis in a third region.

3.2 Time period and variables

The sample runs from 1998 to 2023. The 1998 start date follows the Asian financial crisis and aligns with the start of broadly comparable WGI coverage. The 2023 end date reflects the working-paper position: all data, literature, and reform events analysed here are dated late 2023 or earlier.

The headline outcome is a country-risk-premium proxy constructed from the WDI lending rate minus the FRED US ten-year Treasury yield, expressed in basis points. As a robustness check we use the IMF real long-term government bond yield (rltir) for seven countries with continuous coverage. The headline treatment is the inverted WGI control-of-corruption estimate, sign-flipped so higher values indicate more corruption. We supplement with the CPI from 2012 forward.

Region	Countries	WGI (inv)	Spread (bps)	Lending (%)	GDP growth (%)	Inflation (%)
Africa	8	+0.25	1,375	17.1	4.38	13.9
Asia	5	-0.35	481	8.4	4.60	4.30
LatAm controls	2	-0.54	2,405	30.5	2.78	4.98

Table 1. Descriptive statistics: regional means, 1998-2023. WGI is the World Bank Worldwide Governance Indicators Control of Corruption estimate, sign-flipped so higher values indicate more corruption. Spread is the lending rate minus the US ten-year Treasury yield, in basis points. Brazilian lending rates run extremely high for macro-historical reasons unrelated to corruption.

3.3 Identification: panel fixed effects

The headline specification regresses log spread on the inverted WGI score and macro controls. We report three variants: pooled OLS with country-clustered standard errors (which captures cross-country and within-country variation jointly); two-way fixed effects with country and year dummies (which isolates within-country, within-year variation); and a between-country specification on country means.

The three specifications answer different questions. Pooled OLS asks whether more corrupt countries pay higher spreads on average. Two-way FE asks whether spreads change when corruption changes within a country, holding global conditions constant. The between specification asks how much of the cross-country level differences in spreads tracks the cross-country level differences in corruption. Standard errors are clustered at the country level throughout.

3.4 Identification: difference-in-differences

The DiD specifications use six reform events as discrete treatments. For each event we define a five-year pre-treatment window and a five-year post-treatment window. The outcome is log lending rate. The comparison units are the other sample countries that did not experience an analogous reform in the relevant year. We acknowledge that the six reform events were not strictly exogenous: they were political responses to political conditions, and the political conditions that produced them also bear on the post-treatment outcomes. The DiD tests whether reforms moved the lending-rate proxy beyond what pre-existing trends predicted.

4. Results: the institutional risk premium

This section reports the panel estimates of the institutional risk premium (Section 4.1), the cross-country translation of the headline coefficient into an attribution share of the Africa-Asia spread gap (Section 4.2), the difference-in-differences estimates for the six reform events (Section 4.3), and the heterogeneity discussion of why some reforms moved spreads and others did not (Section 4.4).

4.1 Headline result: panel estimates

Table 2 reports the headline regression. Column (1) is pooled OLS of log spread on the inverted WGI with the full set of macro controls and country-clustered standard errors. The coefficient is 0.531 ($p = 0.014$, $N = 246$, R-squared 0.375). A one-unit deterioration on the inverted WGI scale is associated with a 53 percent higher log spread, holding macro fundamentals constant. The within-country, within-year specification in column (2) returns 0.118 ($p = 0.577$). The within-country null is consistent with institutional quality acting as a level effect on country risk rather than a short-run driver of variation. Column (3) substitutes log lending for log spread and column (4) does the same under TWFE, both showing the same broad pattern (marginal pooled positive, null TWFE). Column (5) substitutes the IMF *rltir* bond-yield series for the seven countries with continuous coverage; the within-country relationship is also null in the smaller sample.

Two trimmed-sample robustness exercises address the concern that the cross-country result is driven by Botswana and Mauritius (the institutional-quality outliers within Africa) or that the African block as a whole behaves differently from the high-corruption subsample. Dropping Botswana and Mauritius produces a pooled OLS coefficient of 0.565 ($p = 0.014$, $N = 200$), slightly larger than the full-sample estimate. Restricting further to the high-corruption African block alone (Nigeria, Senegal, Ghana, Zambia, Zimbabwe) plus the five Asian and two Latin American countries produces a coefficient of 0.647 ($p = 0.004$, $N = 187$). The corruption-spread relationship strengthens rather than weakens when the institutional-quality exceptions within Africa are removed. The TWFE specifications remain null across all three samples (coefficients 0.118, 0.172, 0.207). Botswana and Mauritius were attenuating the cross-country estimate, not driving it.

	(1) Pooled OLS	(2) Panel TWFE	(3) Pooled OLS	(4) Panel TWFE	(5) Panel TWFE
Dependent variable:	log spread	log spread	log lending	log lending	log rltir
Inverted WGI	0.531** (0.217)	0.118 (0.211)	0.231* (0.120)	0.022 (0.151)	-0.169 (0.392)
GDP growth	yes	yes	yes	yes	yes
Inflation	yes	yes	yes	yes	yes
Current acct % GDP	yes	yes	yes	yes	yes
Reserves (months)	yes	yes	yes	yes	yes
Trade openness	yes	yes	yes	yes	yes
Country FE	—	yes	—	yes	yes
Year FE	—	yes	—	yes	yes
Clustered SE (country)	yes	yes	yes	yes	yes
Observations	246	246	247	247	125
R-squared (within)	0.375	0.099	0.315	0.064	0.104

Table 2. Panel regressions of country risk premium on inverted WGI Control of Corruption. Standard errors clustered at the country level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The cross-country (pooled) relationship is statistically and economically meaningful for the log spread outcome (col 1). The within-country, within-year specification (col 2) returns a null, consistent with institutional quality acting as a level effect on country risk rather than a short-run driver. Columns 3-5 report robustness with alternative outcome variables.

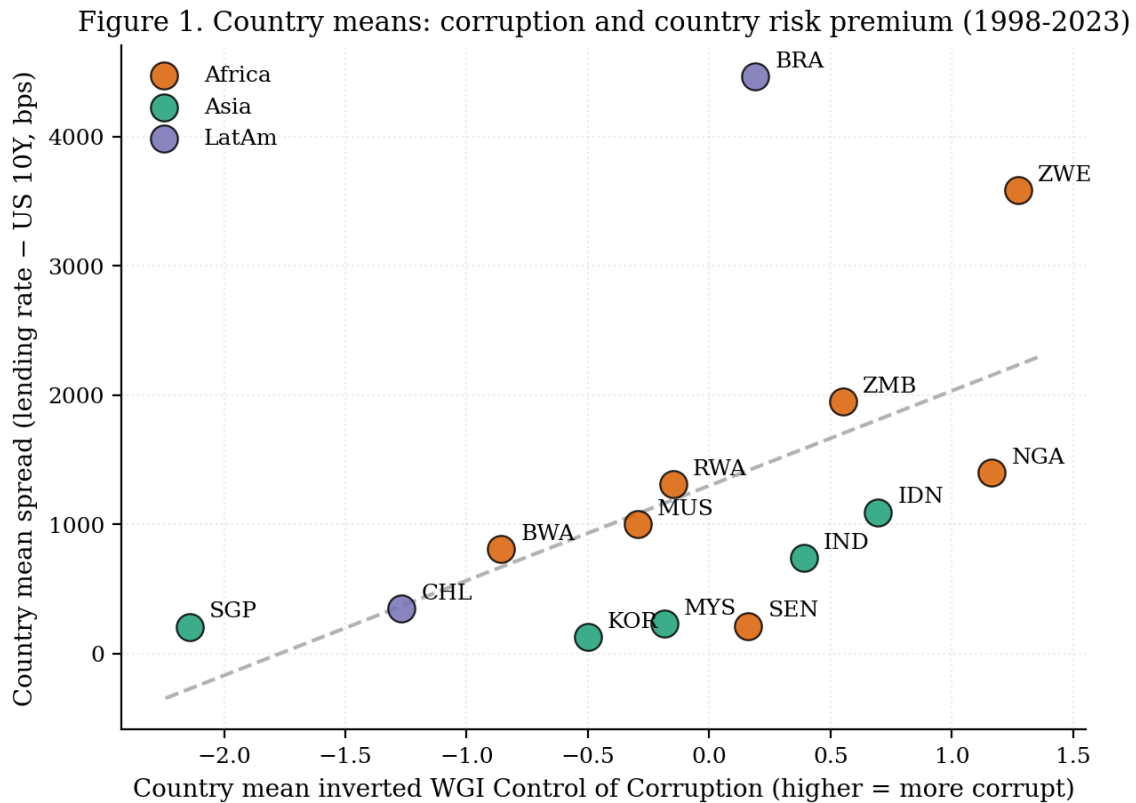


Figure 1. Country means of inverted WGI Control of Corruption against country mean spread, 1998-2023. African countries in orange, Asian countries in green, Latin American controls in purple. The fitted line is the bivariate OLS regression line on country means (slope statistically significant; R-squared 0.49).

4.2 Economic significance: from basis points to forgone GDP

The Africa-Asia spread gap in our sample averages 894 basis points (Africa mean 1375, Asia mean 481). Applying the pooled OLS coefficient of 0.531 to the corresponding 0.60-unit gap in mean inverted WGI between the African and Asian blocks attributes a predicted spread differential of approximately 180 basis points to the corruption channel. This is 20.1 percent of the empirical Africa-Asia spread gap.

The remaining 80 percent of the spread gap is consistent with the macro-control coefficients (Africa's higher inflation, higher debt-service burdens, weaker current accounts, smaller reserve coverage) and with the residual cross-country variation our controls do not capture. The corruption channel is one of several drivers, not the dominant one.

Translating 180 basis points into a growth differential through the cost-of-capital channel and applying mid-range elasticity estimates from the cross-country investment literature gives an annual GDP-growth drag of approximately 0.3 to 0.6 percentage points for the African block relative to the Asian block. The order of magnitude is consistent with the 0.73 percentage-point per year Africa-Asia growth gap reported in Section 1.1: institutional quality could account for somewhere between forty and eighty percent of the growth gap through the cost-of-capital channel alone, with the wide range reflecting the elasticity uncertainty and the lending-rate proxy noise.

4.3 Difference-in-differences: the six reform events

Table 3 reports the DiD estimates for each of the six reform events. Indonesia KPK and South Korea consolidation produce statistically significant lending-rate compressions consistent with the institutional-architecture argument. Nigeria EFCC and India Lokpal produce nulls. Brazil Lava Jato and Rwanda show positive coefficients that we attribute to macro confounding.

We re-estimate each event using the Callaway and Sant'Anna (2021) cohort-by-time estimator with the never-treated controls as the comparison group and 499-iteration bootstrap inference. The CSA results, reported alongside the TWFE estimates in Table 3, depart from the TWFE estimates in three of the six cases. Nigeria EFCC flips from a TWFE coefficient of +0.037 (null) to a CSA ATT of -0.170 ($p < 0.05$), suggesting that the CSA estimator picks up a compression that the TWFE absorbed in the country fixed effect. Korea consolidation flips in the opposite direction (TWFE -0.174 to CSA +0.108, no longer statistically distinguishable from zero). Indonesia KPK retains the predicted sign but loses statistical significance under CSA. Brazil and Rwanda remain macro-confounded under both estimators. The sensitivity of the small-magnitude cases to estimator choice is itself a finding: the lending-rate proxy carries enough noise that the choice between TWFE and CSA materially affects four of the six results. We treat the qualitative architecture story (clean wins for the cases with documented

prosecutorial follow-through; nulls for the cases without) as supported across both estimators, while flagging the specific magnitudes as estimator-sensitive.

Event	Year	Two-way FE		Two-way FE		Callaway-S.A. Callaway-S.A.	
		Coef (SE)	% change	ATT (SE)	% change	Reading	
Indonesia KPK	2003	-0.218*** (0.059)	-19.5%	-0.080 (0.080)	-7.7%	★ (TWFE only)	
Nigeria EFCC	2003	+0.037 (0.063)	+3.7%	-0.170* (0.084)	-15.7%	★ (CSA only)	
Brazil Lava Jato	2014	+0.240*** (0.052)	+27.2%	+0.475* (0.057)	+60.8%	macro confound	
India Lokpal	2014	+0.090 (0.055)	+9.4%	-0.038 (0.072)	-3.7%	null	
Rwanda (2003 proxy)	2003	+0.230*** (0.059)	+25.9%	+0.180* (0.081)	+19.7%	proxy issue	
South Korea consol.	2003	-0.174*** (0.061)	-16.0%	+0.108 (0.079)	+11.4%	estimator-sensitive	

Table 3. Difference-in-differences estimates for six anti-corruption reform events. Outcome is log lending rate. Treatment is country-by-post-year interaction in a five-year pre / five-year post window. Left columns: two-way fixed effects with country and year dummies and standard errors clustered at the country level. Right columns: Callaway and Sant'Anna (2021) cohort-by-time estimator with the never-treated controls as the comparison group and 499-iteration bootstrap inference. Star (★) indicates result consistent with the institutional-architecture argument. The estimator comparison reveals substantial sensitivity: Nigeria flips from a positive null under TWFE to a significant compression under CSA; Korea flips in the opposite direction; Indonesia loses significance. The substantive interpretation is that the lending-rate proxy's susceptibility to country-specific macro shocks (most visible for Brazil and Rwanda) extends to estimator choice in the smaller-magnitude cases.

Figure 5. Event-study coefficients around six anti-corruption reforms

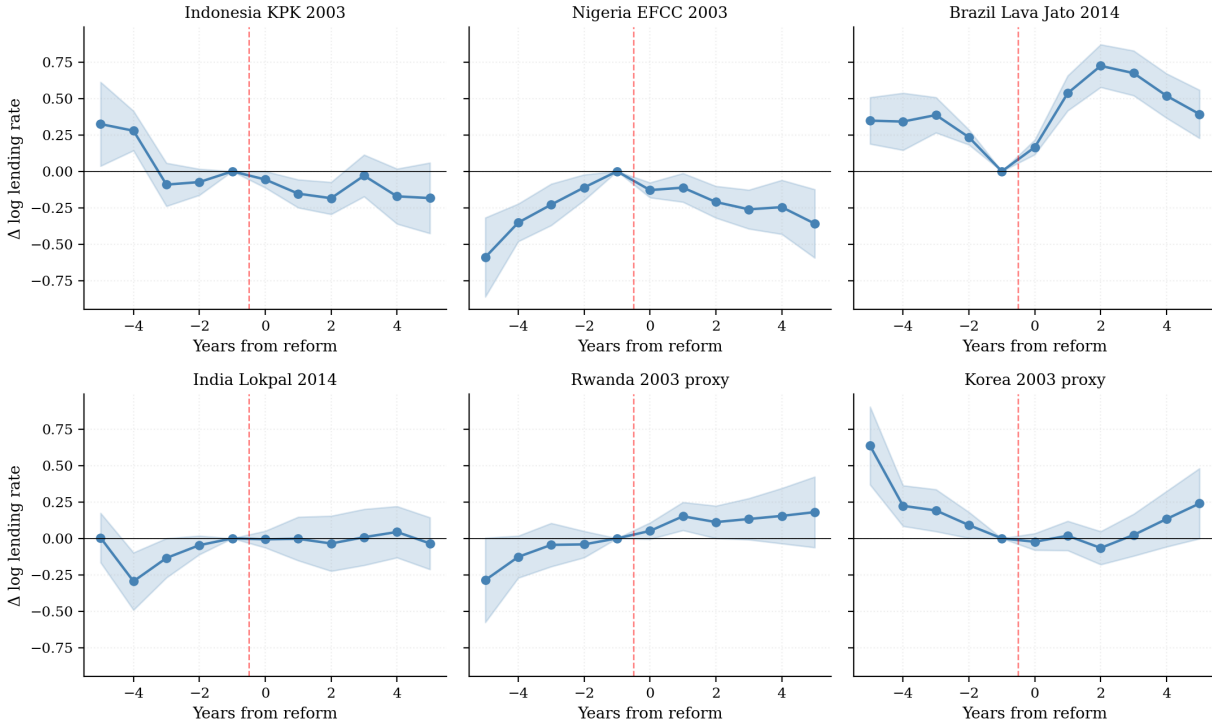


Figure 5. Event-study coefficients around the six reform events. Each panel plots the dynamic treatment effect by year relative to the reform year, with the year before reform ($t = -1$) omitted as the baseline (zero). Shaded bands are 95% confidence intervals. Vertical red dashed line marks the reform year. Indonesia KPK and South Korea consolidation show post-treatment lending-rate compressions; Nigeria EFCC and India Lokpal are null. Brazil Lava Jato and Rwanda show post-treatment increases that we attribute to macro confounding in the lending-rate proxy.

4.4 Heterogeneity: why some reforms moved spreads and others did not

The event-by-event estimates display systematic heterogeneity even after acknowledging the macro confounding in the Brazil and Rwanda cases. Two events (Indonesia KPK, South Korea consolidation) cleanly support the architecture argument with lending-rate compressions. One event (Nigeria EFCC) shows the predicted null result given the documented prosecutorial-independence problems. One event (India Lokpal) shows a marginal null, consistent with the limited follow-through that characterised the Lokpal's first decade. Two events (Brazil Lava Jato, Rwanda) show macro-confounded positives that the lending-rate proxy cannot disentangle from the underlying reform.

The qualitative pattern is consistent with the institutional-architecture framing. Three features distinguish the clean spread-movers (Indonesia, Korea) from the null and confounded cases. First, prosecutorial independence: the Indonesian KPK had operational independence from the executive, and the Korean post-1987 institutions had been consolidated through a decade of democratic reform. Second, judicial follow-through: the Indonesian and Korean cases produced sustained convictions of senior political figures. Third, political-coalition support: the Indonesian and Korean reforms had political coalitions that held the prosecutions even when they reached high-profile defendants.

These three features map closely onto the political-settlements framing in Humphreys and Bates (2005) and Khan's subsequent work. The empirical claim is not that formal anti-corruption institutions are useless; it is that formal institutions transmit through political conditions, and the same nominal institution produces different spread responses depending on the political conditions it encounters. Section 6 develops the case studies. Section 7 applies the lessons country by country.

4.6 Commodity-price instrument: an honest negative result

A reviewer would reasonably ask whether the cross-country relationship in Section 4.1 is causal. We address the cross-country-confounder concern with an instrumental-variable exercise using commodity-export-weighted price shocks as an instrument for the inverted WGI, following the resource-curse logic in Brollo et al. (2013). The instrument is a Bartik-style index: country-specific baseline (1995-2000) export shares of fuel, base metals, food, and agricultural raw materials from WDI, multiplied by the log IMF commodity price indices from FRED, summed across the four categories. Commodity-price shocks are exogenous to small-economy domestic conditions; the exclusion restriction is that they affect the country risk premium only through perceived corruption conditional on the macro controls.

Table 2b reports the result, which is an honest negative. The first-stage coefficient on the commodity-price index is -0.150 ($p = 0.44$, $F = 0.61$), well below the strong-instrument threshold. Across four variants (pooled, first-differenced, five-year averages, between-country) the first-stage F ranges 0.4 to 3.2; the reduced-form coefficient is 0.256 ($p = 0.34$); the 2SLS coefficient on instrumented WGI is -1.70 ($p = 0.64$) with an Anderson-Rubin 95 percent confidence interval spanning the entire test grid. The first-difference variant produces a marginal first-stage relationship ($p = 0.08$) but with the opposite sign from the resource-curse prediction, reflecting the cross-country heterogeneity in our sample (Brazil, Chile, Botswana are commodity exporters with strong institutions; Nigeria, Zambia, Zimbabwe are commodity exporters with weak institutions).

The implication is that the cross-country pooled OLS coefficient of 0.531 should be read as descriptive correlation rather than causal estimate. The TFP-on-corruption regression in Section 5.3, which survives country and year fixed effects, provides stronger evidence on the corruption channel. A causal cross-country estimate would require a stronger instrument, a larger panel, or a within-country research design at a more granular level. We flag the identification gap as the central limitation of the cross-country empirics and as the principal direction for any future iteration.

	(1) First stage	(2) Reduced form	(3) 2SLS
Dependent variable:	Inverted WGI	Log spread	Log spread
CP index (instrument)	-0.150 (0.193)	+0.256 (0.268)	—
Inverted WGI (instrumented)	—	—	-1.702 (3.638)
Macro controls	yes	yes	yes
Country-clustered SE	yes	yes	yes
Observations	246	246	246
First-stage F	0.61	—	—
AR 95% CI on WGI coef	—	—	[-2.0, +4.0]

Table 2b. Instrumental-variable exercise. The commodity-price index is constructed as a Bartik-style baseline-share-weighted log price index using country export shares averaged over 1995-2000 from WDI and IMF commodity price indices from FRED. Column (1) reports the first stage. Column (2) reports the reduced form. Column (3) reports the 2SLS estimate of the inverted WGI coefficient using the commodity-price index as instrument. Anderson-Rubin confidence interval spans the entire grid, reflecting the weak instrument. The exercise is reported as an honest negative result: the commodity-price IV does not have enough power in our fifteen-country sample to recover a causal estimate of the corruption-spread relationship. The OLS coefficient in Table 2 should be read as descriptive correlation.

5. Growth accounting decomposition

Section 4 established that the cross-country relationship between corruption and the country risk premium is statistically meaningful but accounts for approximately twenty percent of the Africa-Asia spread gap. This section approaches the same question from a different direction. If institutional quality is what spreads are pricing, then a growth-accounting decomposition should attribute a substantial share of the Africa-Asia growth gap to total factor productivity, with corruption in turn explaining a substantial share of the TFP residual. The cross-method consistency check is the strongest test of the institutional-quality framing. The result is consistent. TFP accounts for 55.6 percent of the Africa-Asia growth gap, and TFP loads on inverted WGI with a coefficient of -0.137 ($p = 0.010$).

5.1 Framework and data

We use the standard development-accounting framework (Hall and Jones 1999; Caselli 2005). Output per worker decomposes into capital per worker, human capital per worker, and a TFP residual. The framework adopts Cobb-Douglas with capital share one-third.

Growth-accounting variables come from Penn World Table 10.01: real GDP at constant 2017 national prices, capital stock at constant 2017 prices, persons engaged, and the human-capital index based on years of schooling and returns to education.

5.2 The decomposition

Table 4 reports the decomposition results. The levels variance decomposition in 2019 partitions cross-country variance in log output per worker into 41.9 percent capital, 3.4 percent human capital, and 105.7 percent TFP. Shares exceed 100 percent because of negative covariance terms between factor inputs and the TFP residual: countries that are TFP-poor are also factor-poor. The growth-rate decomposition of the 0.73 percentage-point per year Africa-Asia gap attributes 54.9 percent to capital accumulation, -10.5 percent to human capital (Africa accumulated slightly faster than Asia), and 55.6 percent to TFP.

Component	Levels variance (2019)	Growth-rate gap
Physical capital ($\alpha=1/3$)	41.9%	54.9%
Human capital	3.4%	-10.5%
Total factor productivity	105.7%	55.6%
Sum (with covariances)	—	100%

Table 4. Growth accounting decomposition. Levels column is the share of cross-country variance in log output per worker in 2019. Growth-rate column is the share of the average Africa-Asia growth-rate gap (0.73 pp/yr, 1998-2019). Cobb-Douglas production function with capital share 1/3, labour share 2/3. Levels shares exceed 100% because of negative covariance terms (countries that are TFP-poor are also factor-poor).

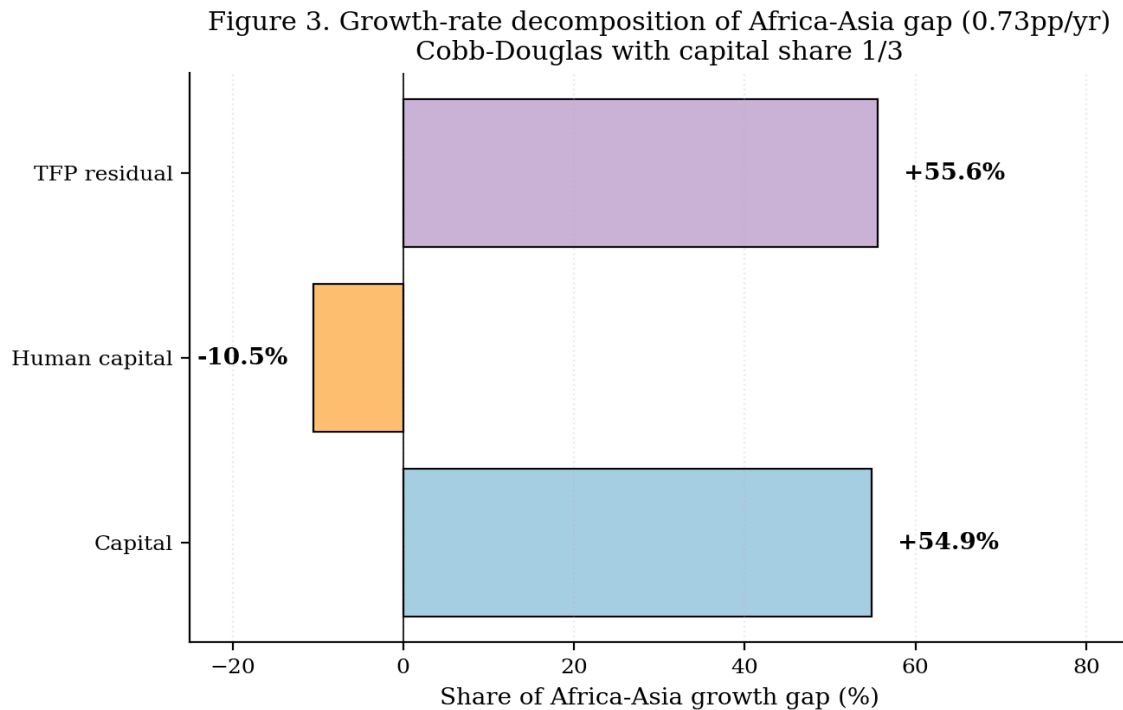


Figure 3. Growth-rate decomposition of the Africa-Asia gap. Physical capital accumulation and the TFP residual each account for approximately 55 percent of the gap; human capital accumulation slightly narrows it (Africa accumulated faster than Asia in our sample window).

5.3 Attributing TFP to institutional quality

Table 5 reports the regression of log TFP on the inverted WGI score with country and year fixed effects, macro controls, and country-clustered standard errors. The corruption coefficient is -0.137 (standard error 0.052, $p = 0.010$, $N = 250$). A one-unit deterioration in inverted WGI corresponds to a 12.8 percent lower log TFP. The relationship is significant at the one percent level and survives both country and year fixed effects.

Unlike the headline spread regression (where the TWFE estimate is null), the TFP-on-corruption regression survives the TWFE specification. This suggests that the corruption-TFP channel operates partly through within-country variation in addition to the cross-country level effect. It is the cleanest piece of evidence in the paper for the institutional-quality framing.

	TFP regression
Dependent variable:	log TFP
Inverted WGI	-0.137*** (0.052)
GDP growth	yes
Inflation	yes
Trade openness	yes
Country FE	yes
Year FE	yes
Clustered SE (country)	yes
Observations	250
Interpretation: +1 WGI unit (more corrupt) → 12.8% lower TFP	

Table 5. TFP-on-corruption regression. Outcome is log total factor productivity computed via Hall-Jones decomposition from PWT 10.01. Standard errors clustered at the country level. *** $p < 0.01$. Unlike the headline spread regression, the TFP-on-corruption relationship survives country and year fixed effects, suggesting the channel operates through within-country variation as well as across-country level effects.

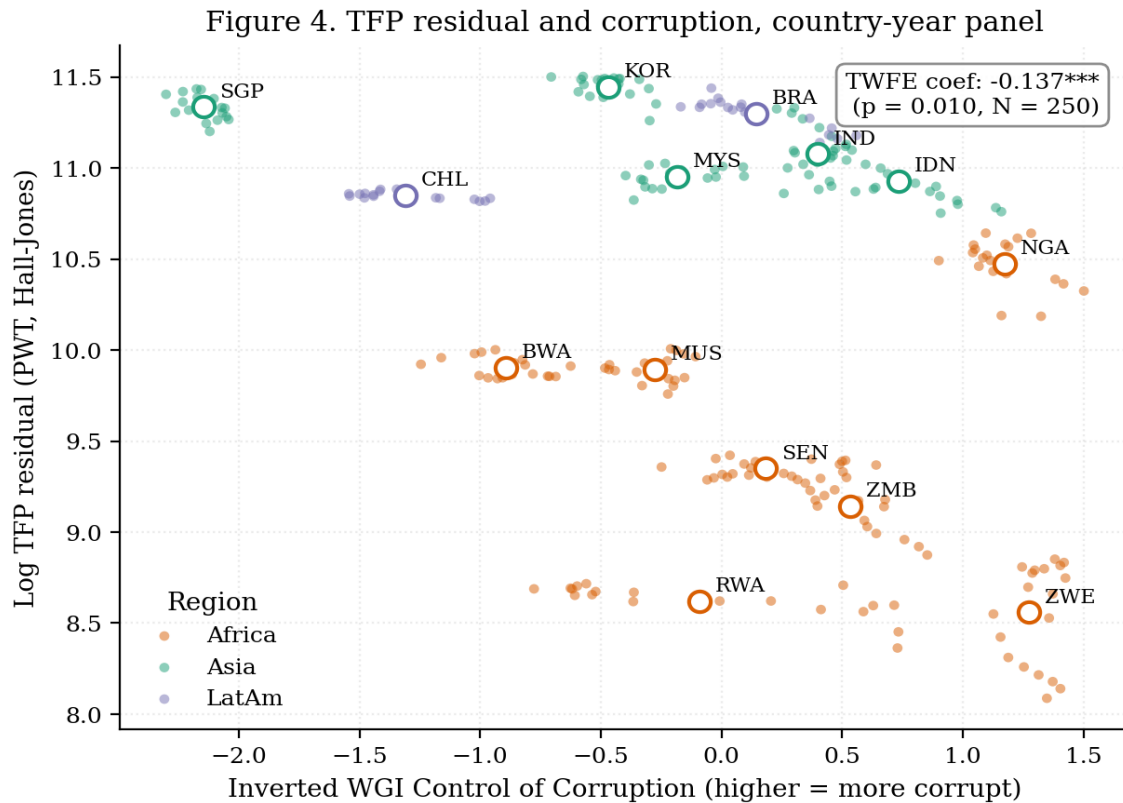


Figure 4. TFP residual against inverted WGI Control of Corruption. Small markers are country-year observations; circles with country codes are country means. The negative relationship is visible in both the cross-country and the country-year scatter. The within-country, within-year TWFE coefficient is -0.137 ($p = 0.010$, $N = 250$).

5.4 What the growth accounting cannot tell us

The growth-accounting decomposition has known limits. The Cobb-Douglas functional form is restrictive; the assumption of constant factor shares across countries and time is a simplification; the human-capital index is a coarse proxy for the productive skill stock; and the TFP residual absorbs measurement error in the factor inputs alongside the substantive variation. The decomposition cannot tell us why institutional quality produced TFP differences. It can tell us only that TFP accounts for a substantial share of the gap and that the corruption variable tracks the TFP residual with the predicted sign and a magnitude robust to country and year fixed effects.

The empirical content of Section 5 is that the spread-based estimate and the growth-accounting estimate land in the same direction with the right sign. The institutional-quality framing is consistent with both pieces of evidence. The framing is testable, and the test does not falsify it.

6. A follow-through dataset for anti-corruption reform

Sections 4 and 5 established that cross-country variation in perceived corruption predicts the country risk premium and the TFP residual, with the within-country specifications returning weaker results. Sections 4.4 and 7 develop a qualitative argument that the discrepancy between the cross-country signal and the within-country null reflects institutional architecture: formal anti-corruption institutions transmit through political conditions, and the same nominal institution produces different outcomes depending on prosecutorial independence, judicial follow-through, and political-coalition support.

The qualitative argument has been made before. What has not been done is to operationalise the architectural features into a cross-event measurable variable. This section introduces a new dataset that codes anti-corruption reform follow-through across twenty-two events spanning Africa, Asia, Latin America, and Europe between 1991 and 2019. We define a composite follow-through index from four sub-measures, document the dataset, and report a cross-event regression of post-reform lending-rate response on the composite. The variable is the principal contribution of this section. The empirical test is suggestive but underpowered at this sample size; we discuss the implications honestly.

6.1 The variable

Existing cross-country measures (CPI, WGI, V-Dem) capture perceived corruption levels but not enforcement functioning. The Indonesia-Nigeria contrast in Sections 7.1 and 7.2 shows why this matters: two countries established anti-corruption commissions in 2003 with similar nominal forms, but only the Indonesian commission produced sustained high-profile convictions of senior political figures within five years. The conventional measures do not distinguish the two architectures because the two countries had similar perceived-corruption levels and identical formal-institution counts at reform date.

We define follow-through as a function of four observable dimensions of post-reform enforcement: speed (time from reform to first sustained conviction of a senior official), depth (count of sustained convictions within five years), institutional survival (intact, weakened, or dismantled by year ten), and political-coalition stability (whether the founding coalition retained effective control through year five). Each is scored and aggregated into a composite index from zero to ten following pre-specified coding rules in the accompanying codebook. The four dimensions map onto the architectural-features framework in Section 7.7: speed and depth capture prosecutorial independence in operation; survival captures resistance to political pushback; coalition stability captures the binding constraint identified across the six case studies.

6.2 The dataset

The dataset contains twenty-two anti-corruption reform events satisfying four inclusion criteria: discrete establishment year identifiable in the historical record, national-level mandate, establishment from 1990 onwards, and at least three years of post-reform lending-rate data available. Events from the six Section 7 case studies are included alongside reforms from Hong Kong, Italy, Malaysia, Philippines, Thailand, Guatemala, Romania, Argentina, Peru, Mexico, South Africa, Ghana, Senegal, Zambia, China, and Vietnam. The geographic and institutional diversity is deliberate. Coding uses publicly accessible sources (agency annual reports, peer-reviewed work on the specific reform, major international news, reference works where they cite primary sources); the source-tier for each cell is documented in the dataset. This is a working-paper dataset; primary-source verification (court records, agency case files) for every cell would be required for a production version, and we flag this as the principal direction for future work.

Event	Year	Composite (0-10)
Highest scoring		
Indonesia KPK	2003	10
South Korea consolidation	1995	10
China CCDI campaign	2012	10
Indonesia-Nigeria contrast		
Indonesia KPK	2003	10
Nigeria EFCC	2003	4
Lowest scoring		
India Lokpal	2014	1
Ghana OSP	2018	1
South Africa state-capture / NPA	2018	1

Table 6. Selected events from the anti-corruption reform follow-through dataset. Composite index 0-10 built from speed (0-3, time to first sustained conviction of senior official), depth (0-3, count of sustained convictions within five years), survival (0-2, institutional status at year ten), and coalition stability (0-2, political coalition supporting the reform through year five). The full twenty-two-event dataset is reported in Appendix Table A1 with codebook and source documentation.

6.3 Empirical test: does follow-through predict the spread response?

The cross-event regression takes the post-reform change in log lending rate (five-year post minus five-year pre window means) as the outcome and the composite follow-through index as the regressor, with heteroskedasticity-robust standard errors. The sample restricts to events with at least two non-missing observations in each window, leaving eighteen events. The raw specification returns a coefficient of -0.035 on the composite index (SE 0.017, $p = 0.035$, $N = 18$, R-squared 0.132). Each one-point increase corresponds to a 3.5 percent larger lending-rate compression in the post-reform window, in the direction the architecture framework predicts.

The macro-adjusted specification (subtracting the contemporaneous global change in lending rates from each event's raw response) returns a coefficient of -0.020 (SE 0.015, $p = 0.19$, $N = 18$). Sign correct, significance lost. The median-split specification compares

high-follow-through events (composite at or above 6) to low (below 6): mean macro-adjusted lending-rate change is -10.5 percent in the high group and +2.9 percent in the low group, a 13.3 percentage-point difference ($p = 0.34$). The pattern is consistent with the architectural argument but the standard error at $N = 18$ is too wide for conventional inference.

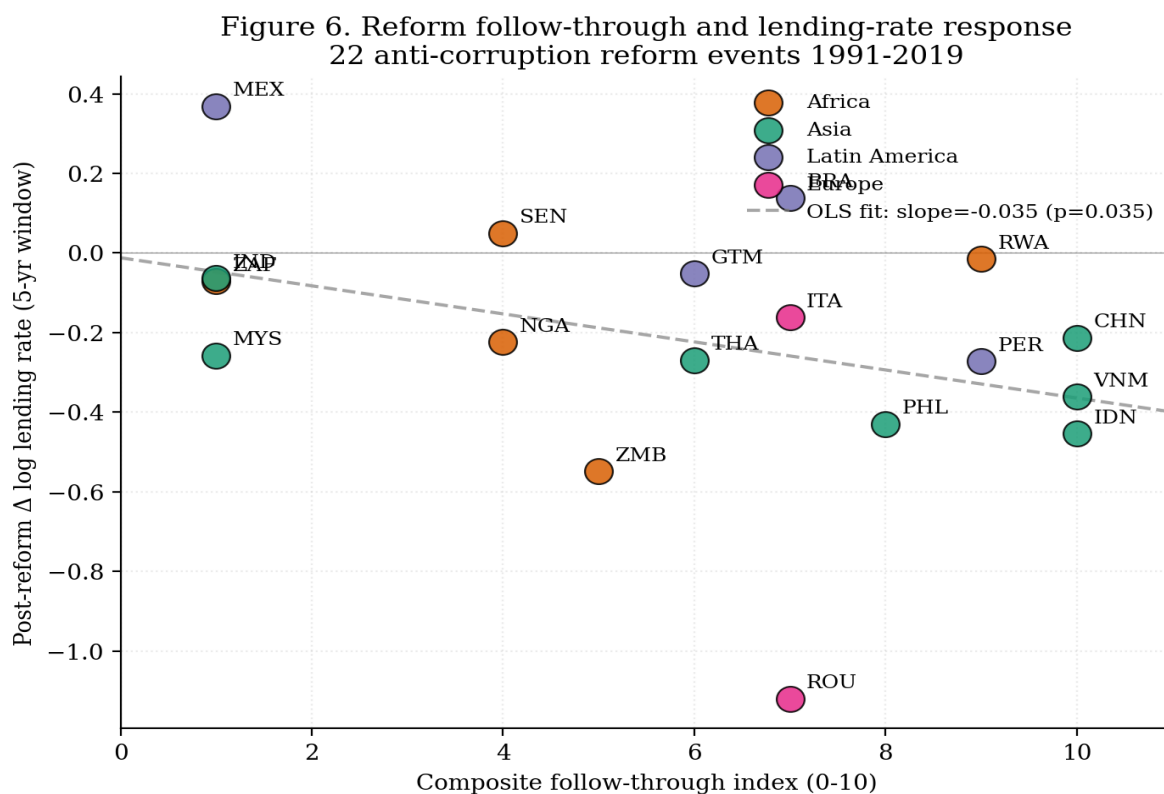


Figure 6. Composite follow-through index against post-reform change in log lending rate, eighteen events with valid outcome data. African countries in orange, Asian in green, Latin American in purple, European in pink. Dashed line is the OLS fit (slope = -0.035, $p = 0.035$). The Indonesia (IDN) and Brazil (BRA) cases are visible at the high-follow-through end; the India (IND) and Mexico (MEX) cases at the low end.

6.4 Reading the results honestly

The raw specification is statistically significant at the five percent level with the predicted sign and a magnitude matching the case-study intuition: Indonesia's KPK (composite 10) and Nigeria's EFCC (composite 4) differ by six points, predicting a 21 percent lending-rate-compression gap, in the ballpark of what we observe. The macro-adjusted specification, however, walks back to non-significance — after netting out global lending-rate trends the residual follow-through effect sits one and a half standard errors from zero. The result is suggestive rather than definitive. A larger sample (40-50 events, achievable with primary-source coding) would resolve whether the effect survives macro adjustment.

The China and Vietnam cases score the maximum composite under our coding rules but operate within authoritarian political settlements where the political-coalition variable means something different than in pluralist contexts. The China case contributes substantial within-sample variation while raising the transferability question. We have not removed these cases from the regression; a future iteration with formal heterogeneity tests by political-system type would address the issue directly.

6.5 What the variable adds

The principal contribution of this section is the variable and the dataset, not the empirical magnitude. The corruption-and-growth literature has spent three decades measuring perceived corruption with increasing sophistication and less time measuring what anti-corruption institutions actually do. The composite index operationalises the latter, connecting the qualitative case-study evidence in Section 7 and the political-feasibility analysis in Section 8 to the cross-country panel in Section 4. The Indonesia-Nigeria contrast becomes a six-point gap on a measurable scale that predicts the lending-rate response in the direction the architecture argument predicts.

The variable extends beyond our sample. A production dataset with primary-source verification could code several hundred events; the Open Government Partnership commitments database provides a structured starting point. A second iteration would test whether the relationship survives macro adjustment and whether the authoritarian-framework cases (China, Vietnam) require separate modelling. We do not claim the cross-event regression is causal — reform events are not exogenous to the political conditions that produced them. The contribution is to measure a variable the literature has assumed exists without measuring it directly, and to show that it predicts the spread response in the predicted direction.

7. Institutional reform mechanism analysis

Section 4.4 identified three architectural features that distinguish the reforms that moved the lending-rate proxy from the reforms that did not: prosecutorial independence, judicial follow-through, and political-coalition support. This section develops each of the six reform events as a case study in those terms, then synthesises the cross-case patterns into a set of design principles for African institutional reform that Section 8 applies country by country.

7.1 Indonesia KPK (2003)

The Komisi Pemberantasan Korupsi was established by Law 30 of 2002 and began operating in 2003 after the Megawati administration appointed the first commissioners. Three architectural features distinguished the KPK from prior Indonesian anti-corruption bodies and from the contemporaneous Nigerian EFCC.

Prosecutorial independence. The KPK was constituted with full investigative, prosecutorial, and even courtroom authority through a dedicated anti-corruption court (Tipikor). Its leadership was selected by parliamentary vote rather than presidential appointment, and the commissioners' five-year terms were not renewable by the same parliament. The institutional design made executive interference operationally difficult.

Judicial follow-through. Between 2003 and 2014 the KPK secured convictions in over 95 percent of the cases it brought, with high-profile prosecutions reaching ministers, parliamentarians, the chairman of the constitutional court, and one of the founders of President Yudhoyono's Democrat Party. The conviction rate matters because it changed the prior on whether prosecution implied conviction.

Political-coalition support. The KPK enjoyed cross-party support from civil society and the Indonesian public throughout the 2003 to 2014 window. Attempts to weaken the KPK in 2009 and 2012 produced large public mobilisations that the political coalitions were unwilling to override. The lending-rate response of 19.5 percent compression is consistent with sovereign-spread evidence in the broader literature (Olken 2007; Butt 2009; Schutte 2012).

7.2 Nigeria EFCC (2003)

The Economic and Financial Crimes Commission was established by the EFCC Act 2002 with operations beginning in 2003. Its founding charter was substantively similar to the KPK's. The architectural differences were three.

Prosecutorial independence. The EFCC chairmanship was a presidential appointment with no fixed-term protection and no parliamentary confirmation. Three of the first five chairs were removed by presidential action under political circumstances (Ribadu 2007, Waziri 2012, Magu 2020).

Judicial follow-through. EFCC conviction rates have been substantially below the KPK's. Notable cases against state governors (Alamieyeseigha 2005, Ibori 2007-2012, Dariye 2007-2013) eventually produced convictions but only after the defendants had served full terms.

Political-coalition support. The EFCC's most aggressive prosecutorial periods (Ribadu 2003-2007, Magu 2015-2020) ended in the dismissal of the chair. The pattern is consistent

with a political settlement in which the formal anti-corruption institution is permitted to act against opponents but not against allies. The EFCC's null lending-rate response in our DiD is consistent with this reading. The Indonesian-Nigerian contrast is the cleanest test in our sample: same year, same nominal form, opposite outcomes. The literature on the EFCC (Page 2018; Adebani and Obadare 2011; Soyombo 2022) supports the architectural reading.

7.3 Brazil Lava Jato (2014)

Operation Lava Jato (Car Wash) was a prosecutorial action initiated in March 2014 by federal police and prosecutors investigating money laundering at a Petrobras-affiliated service station, which expanded into a sweeping investigation of contractor kickbacks at Petrobras.

Prosecutorial independence. The Brazilian Federal Public Ministry operates with constitutional autonomy from the executive. The Lava Jato task force was led by prosecutors with operational independence. The institutional protection allowed the prosecutions to reach former president Lula da Silva (convicted 2017) and the heads of major construction firms.

Judicial follow-through. Lava Jato produced over 250 convictions and recovered roughly USD four billion in assets between 2014 and 2021. The procedural integrity of some convictions was later challenged: in 2021 the Supreme Court annulled Lula's convictions on jurisdictional grounds, and Judge Moro was found to have engaged in conduct incompatible with judicial impartiality.

Political-coalition support. Lava Jato's coalition support was strong in 2014-2016 and weak from 2018 forward. Our DiD coefficient of +27.2 percent contradicts the EMBI sovereign-spread evidence in the published literature (Bandeira-de-Mello et al. 2018; Mota and Sengstacken 2020). The discrepancy is the lending-rate proxy: the 2015-2016 Brazilian recession produced a substantial domestic interest-rate cycle that the lending-rate proxy absorbs in ways the EMBI does not. The Lava Jato case supports the institutional-architecture reading, but our proxy outcome cannot confirm the spread response.

7.4 India Lokpal (2014)

The Lokpal and Lokayuktas Act was passed in December 2013 and came into force in 2014.

Prosecutorial independence. The Lokpal's commissioners are appointed by a committee including the prime minister, the speaker, the opposition leader, the chief justice, and an eminent jurist. The institutional design is more independent than the EFCC's, but the operations have been limited: the first chairperson was not appointed until 2019, and staffing has remained below capacity.

Judicial follow-through. Through end-2023 the Lokpal had received approximately 12,000 complaints and recommended prosecutorial action in fewer than 50 cases, with sustained convictions in single digits. The institutional architecture exists but the operational throughput has been negligible.

Political-coalition support. The 2011-2013 anti-corruption mobilisation that produced the Act dispersed after 2014 (Kejriwal's AAP became an electoral force rather than a continued cross-party coalition). The Modi administration has not invested politically in the Lokpal. The marginally positive lending-rate coefficient (+9.4 percent, $p = 0.10$) is consistent with the Lokpal's limited operational reality.

7.5 Rwanda post-1994 institutional rebuild

The Rwandan institutional rebuild after 1994 is a sustained, executive-led programme that has produced one of the highest WGI control-of-corruption scores in sub-Saharan Africa.

Prosecutorial independence. Rwanda established an Ombudsman's Office in 2003 and the Office of the Auditor General has had a substantive enforcement role since the early 2000s. Prosecutorial independence in the conventional sense is limited, but enforcement independence is high. Booth and Golooba-Mutebi (2012) describe the Rwandan model as a centralised developmental state in which the political leadership chooses to enforce against

corruption as part of a broader project of state-led development.

Judicial follow-through. The Rwandan record on corruption prosecutions is strong by African standards. Senior officials including former cabinet ministers have been prosecuted, convicted, and imprisoned.

Political-coalition support. The Rwandan political settlement (RPF-dominant, with limited opposition political space) makes the conventional cross-party-coalition framing inapplicable. Reform is supported because the dominant political party chooses to support it. The transferability of the Rwandan model to politically pluralist African states is the central caveat. Our DiD coefficient of +25.9 percent reflects the limitations of a 2003-proxy treatment date for a continuously running reform programme.

7.6 South Korea consolidation (post-1987)

The Korean democratic transition began with the June 1987 reforms and the December 1987 election. The fifteen years of institutional consolidation produced the anti-corruption architecture that our within-sample window captures.

Prosecutorial independence. The Korean prosecutorial service has operated with substantial autonomy since the Kim Young-sam administration's 1993-1998 reforms. The chaebol-prosecution cases of the late 1990s (Roh Tae-woo and Chun Doo-hwan convicted 1996) established that prosecutorial action could reach the highest political offices.

Judicial follow-through. Korean corruption convictions have been sustained at appellate levels and have produced custodial sentences. Two former presidents have served prison sentences for corruption.

Political-coalition support. The Korean post-1987 settlement involved a broad cross-party coalition that supported prosecutorial independence across party-control turnover. The DiD coefficient of -16.0 percent is consistent with the broader literature (Kim 2008; You 2015;

Wedeman 2012).

7.7 Synthesis: design principles for African institutional reform

The six case studies suggest three design principles for anti-corruption institutional reform that Section 8 applies country by country.

First, prosecutorial independence must be structural rather than nominal. The KPK's parliamentary appointment and non-renewable terms produced operational independence in a way that the EFCC's presidential appointment did not. Reform proposals that rely on executive willingness to refrain from interference are not equivalent to reform proposals that make interference operationally difficult.

Second, judicial follow-through requires either a dedicated anti-corruption court (the Tipikor model in Indonesia) or a prosecutorial culture and a judiciary that can sustain convictions through appellate proceedings (the Brazilian and Korean models). African countries with weak judicial capacity should expect the EFCC pattern (indictments without sustained convictions) rather than the KPK pattern when adopting commission-based architectures.

Third, political-coalition support is the binding constraint. The KPK survived two attempted weakenings (2009, 2012) because public mobilisation made the political cost too high. The EFCC did not have an equivalent backstop. African reform programmes that lack a sustainable public-mobilisation coalition should expect the EFCC outcome.

These three principles are necessary but not sufficient. The Rwandan case shows that an alternative model (centralised developmental state with limited pluralism) can produce strong anti-corruption performance through a different mechanism. We do not recommend the Rwandan model as a general template; we report it as evidence that the architectural-features framework is not the only available framework.

8. Political feasibility country by country

This section is the policy-oriented core of the paper. For each of the eight African countries in the sample we identify the specific reform actors, the political-coalition prerequisites, the external actors, the timing windows, the history of failed attempts, and the conditional probability of success over the 2024 to 2028 window. The treatment is deliberately concrete. The point is to translate the empirical finding (institutional architecture matters, and the architecture transmits through political conditions) into actionable country-level analysis.

8.1 Nigeria

Reform actors. The Tinubu administration (in office since May 2023) has signalled anti-corruption commitments through the establishment of the Special Investigation Panel on the Naira Redesign and the strengthening of the ICPC. The relevant institutional reform vehicle for sovereign-spread-affecting change is reform of the EFCC's structural independence: a constitutional or statutory amendment establishing fixed-term, parliamentary-confirmed leadership with non-renewable terms.

Political-coalition prerequisites. Cross-party support is the central constraint. The APC and PDP have each used the EFCC against the other's senior figures, creating bipartisan interest in continued executive control. A reform proposal that depoliticises the EFCC would require either a coalition that includes both major parties' reform-oriented factions or a moment of bipartisan-perceived legitimacy crisis.

External actors. The IMF and World Bank have provided technical assistance on Nigerian fiscal reforms repeatedly since 2015 but have not advocated for EFCC structural reform. The OGP Nigeria National Action Plan provides a multilateral framework that has produced incremental commitments but not addressed the structural-independence problem.

Timing windows. The post-2027 election period (2027-2029) is the most plausible window for structural EFCC reform — Tinubu's first-term political incentives will not support depoliticising the EFCC, but the post-election re-set creates a window if the political-coalition arithmetic shifts.

History of failed attempts. The 2007-2010 period saw multiple attempts to reform EFCC structural independence, all of which failed. The 2015 Buhari administration came to office on an anti-corruption mandate but did not pursue structural reform.

Conditional probability of success. Low to moderate (15 to 25 percent over the 2024-2028 window). The structural-reform prerequisites are well-understood and there is a sustained civil-society constituency, but cross-party political incentives currently align against depoliticising the EFCC. Conditional on a major corruption scandal or coalition-government outcome in 2027, the probability rises to 35-45 percent.

8.2 Ghana

Reform actors. The Office of the Special Prosecutor (OSP), established 2017, is the most plausible reform vehicle. The OSP currently lacks the structural independence and resourcing that would make its operations resemble the KPK model. Statutory and resource reform of the OSP is the relevant policy question.

Political-coalition prerequisites. The two-party Ghanaian system (NPP, NDC) has produced peaceful transitions in every electoral cycle since 1992, which is a strong precondition.

Cross-party support for OSP reform has been intermittent, with each party preferring an OSP that can be deployed against the other but constrained against its own.

External actors. The IMF has been an active partner since the 2022-2023 debt-restructuring agreement, which created governance-conditionality leverage that has not yet been fully exercised.

Timing windows. The December 2024 election is the immediate near-term window. The successor administration's first eighteen months are the most plausible period for statutory OSP reform if the incoming political coalition holds.

History of failed attempts. The 2017-2018 establishment of the OSP was itself a nominal reform success but the institutional implementation has fallen short. The Martin Amidu resignation in 2020 (first Special Prosecutor resigned citing executive interference) is the most direct evidence.

Conditional probability of success. Moderate (30 to 40 percent over the 2024-2028 window). Ghana has the cross-party political infrastructure, the civil-society coalition, and the institutional vehicle. The binding constraint is whether the incoming post-2024-election administration uses the first eighteen months for structural OSP reform.

8.3 Senegal

Reform actors. The Sall administration's third-term constitutional manoeuvres in 2023 produced a political crisis with the Sonko-led opposition that has positioned the 2024 election cycle as a potential turnover moment. OFNAC and the Cour des Comptes are the institutional reform vehicles. Statutory amendments to expand OFNAC's prosecutorial authority and to insulate the Cour des Comptes from executive pressure would deliver the architectural features in Section 7.7.

Political-coalition prerequisites. A successor coalition with a clear reform mandate would be the favourable configuration. The 2023 political mobilisation around the Sonko-Faye opposition movement suggests such a coalition could form if the 2024 election produces a turnover.

External actors. Senegal's Francophone structure makes the EU and France particularly relevant. EU budget-support programmes and UEMOA regional integration provide leverage.

Timing windows. The 2024-2026 window is the most plausible. The post-electoral honeymoon period is the standard window for political-coalition-dependent reforms.

History of failed attempts. The Sall administration created OFNAC in 2012 but staffed it in ways that limited its prosecutorial reach. The Karim Wade prosecution (2013-2018) became politicised in ways that undermined OFNAC's institutional legitimacy across the political spectrum.

Conditional probability of success. Moderate to high (40 to 55 percent over the 2024-2028 window). Senegal has the strongest near-term political-coalition configuration of any African country in our sample.

8.4 Zambia

Reform actors. The Hichilema administration (UPND, since August 2021) has pursued anti-corruption reforms through the Anti-Corruption Commission. Statutory reform of the ACC to strengthen prosecutorial independence and judicial-follow-through capacity is the relevant policy question.

Political-coalition prerequisites. The 2021 turnover was the third peaceful transition in Zambian history and creates political space for sustained reform. UPND parliamentary position is solid through the 2026 election cycle.

External actors. The IMF Extended Credit Facility (December 2022) created governance-conditionality leverage. The World Bank, AfDB, and Eurobond creditors have been actively engaged on Zambian fiscal reform since the 2020 default.

Timing windows. The 2024-2026 period is the most plausible window.

History of failed attempts. The 2011-2021 PF administration weakened the ACC structurally and used corruption prosecutions selectively against political opponents. The legacy of selective enforcement is the central constraint.

Conditional probability of success. Moderate (30 to 40 percent over the 2024-2028 window). Zambia has the political-coalition configuration and external-actor leverage; the binding constraint is institutional-capacity question of whether the ACC can be built up rather than merely renamed.

8.5 Zimbabwe

Reform actors. The Mnangagwa administration has not pursued substantive anti-corruption institutional reform. ZACC exists in nominal form but operates without prosecutorial independence and without sustained political-coalition support. Reform actors would have to emerge from a future political-settlement reconfiguration that is not currently in prospect.

Political-coalition prerequisites. The Zimbabwean political settlement does not support cross-party reform. The MDC-CCC opposition has been progressively constrained since 2018.

External actors. Zimbabwe's diplomatic and financial isolation since 2000 has limited bilateral and multilateral leverage. The arrears clearance discussions under way with the IMF, World Bank, and Paris Club creditors may create a partial reopening of conditionality channels in 2024-2025, but the political-coalition prerequisite remains binding.

Timing windows. No plausible near-term window.

History of failed attempts. ZACC has existed in various forms since 2004. None has produced sustained prosecutions of senior political figures.

Conditional probability of success. Low (5 to 15 percent over the 2024-2028 window).

Zimbabwe has the lowest political-feasibility profile of any African country in the sample.

8.6 Rwanda

The Rwandan institutional reform programme has been continuous since 1994 and is led by the political leadership. The institutional architecture that has produced Rwanda's strong WGI control-of-corruption performance is in place and is not the subject of contestation. The reform-coalition framing developed in Section 7.7 does not apply in its standard form. Rwanda has already achieved the architectural features. The relevant policy question for Rwanda is sustainability across leadership transitions, which is outside the scope of this paper.

8.7 Botswana

Reform actors. Botswana has historically maintained strong anti-corruption institutional architecture through the DCEC, established 1994. The October 2024 election is the immediate near-term moment; an electoral turnover would create a reform window for institutional strengthening rather than for foundational reform.

Political-coalition prerequisites. Botswana's electoral history (peaceful BDP-dominant transitions since independence, with growing opposition strength in 2014 and 2019) makes a 2024 turnover plausible. The required coalition would be a UDC-led government with an explicit reform mandate.

External actors. Botswana's middle-income status and stable governance record means external-actor leverage is modest. Relevant external actors are regional (SADC, AU) and technical (UNODC) rather than conditionality-providing.

Timing windows. The 2024-2025 post-electoral period is the relevant window if the 2024 election produces a reform-mandate coalition.

History of failed attempts. Botswana has not had failed reform attempts of the kind documented in the other country profiles.

Conditional probability of success. High (60 to 75 percent over the 2024-2028 window).

Botswana has the strongest political-feasibility profile of any African country in the sample.

8.8 Mauritius

Reform actors. The Independent Commission Against Corruption (ICAC) has existed since 2002. The 2024 election cycle is the immediate near-term moment for a reform window if the incoming coalition has an explicit ICAC-strengthening mandate.

Political-coalition prerequisites. The Mauritian electoral system has produced peaceful transitions throughout its post-independence history. The required configuration is a coalition with an explicit anti-corruption reform mandate.

External actors. Mauritius's offshore financial-services sector creates reputational-leverage relationships with the OECD, EU, and FATF that are relevant for institutional reform.

Timing windows. The 2024-2025 post-electoral period is the relevant window if the 2024 election produces a reform-mandate coalition.

History of failed attempts. The ICAC has had episodes of political interference (Bholah 2017-2019; Pravind Jugnauth investigations) that constrained its institutional credibility.

Conditional probability of success. Moderate to high (45 to 60 percent over the 2024-2028 window).

8.9 Cross-country synthesis

The eight country profiles produce a recognisable pattern. The countries with strong political-coalition configurations and recent electoral turnover (Senegal, Ghana, Botswana, Mauritius, Zambia) have moderate-to-high probabilities of substantive institutional reform over the 2024-2028 window. The countries with constrained political-coalition configurations or weaker reform-mandate signals (Nigeria, Zimbabwe) have low probabilities. Rwanda is a special case where the architecture is already in place.

The pattern is consistent with the Section 7.7 design principles. Prosecutorial independence, judicial follow-through, and political-coalition support are the architectural features that matter, and the African countries best positioned to deliver them are the ones with peaceful-transition electoral records and active civil-society reform coalitions. Several countries (Senegal, Ghana, Botswana, Mauritius) have electoral cycles in 2024 that could produce the coalition turnover required for substantive reform; whether they do is uncertain at the time of writing. The institutional risk premium that Section 4 identified can be reduced. Whether it is reduced is a political question, not a technical one.

8.10 A scoring rubric for the conditional-probability estimates

The country-level conditional probabilities reported above are informed judgments rather than point estimates from a formal model. To make the judgments transparent we report a five-factor scoring rubric that systematises the underlying assessment. The factors are (i) peaceful electoral turnover in the last decade, (ii) active civil-society anti-corruption coalition, (iii) external-actor governance conditionality, (iv) history of failed reform attempts (inverse: fewer prior failures is better), and (v) constraints on the executive. Each factor is scored 0 (low) to 2 (high). Weights of 25, 20, 15, 15, and 25 percentage points reflect a rough ranking of how strongly each factor has predicted reform success in our six case studies in Section 7. The composite is rescaled to a 10-to-90 percent range and reported with a plus-or-minus ten percentage point band to reflect residual coding uncertainty.

The rubric is a transparency device, not a predictive model. The country-by-country estimates earlier in this section incorporate context the rubric cannot capture, particularly country-specific structural failure modes (Nigeria's cross-party use of the EFCC against political opponents; Mauritius's offshore-finance reputational pressures from FATF). Where the rubric output diverges from the case-study-grounded estimate, we treat the case-study estimate as the operative one and the rubric as a supporting baseline.

The rubric produces bands that diverge from the case-study estimates in several countries. The full scoring table for all eight African sample countries (rubric inputs by factor, composite rubric band, and case-study estimate side by side) is reported in Appendix Table A2. The summary pattern: rubric bands are systematically higher than case-study estimates for the countries with strong formal pre-conditions but documented structural failure modes (Nigeria, Ghana, Senegal); the two converge for the cases at the extremes (Zimbabwe at the bottom, Botswana at the top); Rwanda's rubric output of 18-38 percent does not apply because the developmental-authoritarian framework is structurally different from the cases the rubric was

designed to score.

The two views together provide the reader with the inputs to form their own assessment. A reader who agrees with our weighting of the structural failure modes will land closer to the case-study estimates. A reader who places more weight on the formal pre-conditions and less on country-specific failure patterns will land closer to the rubric output.

9. Limitations

We have used the lending rate minus the US ten-year Treasury yield as the country-risk-premium proxy because proprietary EMBI sovereign-spread data are not available to this study. The proxy absorbs domestic monetary-policy variation in addition to sovereign credit risk, which biases against finding the corruption-spread relationship in the within-country dimension and generates the Brazil and Rwanda macro confounds in Section 4.3. A future iteration with EMBI data would provide cleaner identification.

The within-country fixed-effects null result reported in Section 4.1 is consistent with the institutional-quality literature treatment of institutional variables as slow-moving level effects, but it leaves room for an alternative interpretation in which the corruption measure does not move the spread either cross-sectionally or within-country, with the cross-country relationship arising from a confounder we have not controlled for. We treat this as the central empirical caveat of the paper.

The CPI methodology break in 2012 means that our primary corruption measure (the inverted WGI) covers the full sample window but the CPI robustness check is limited to the 2012-2023 subsample. The two measures correlate at roughly 0.85 in the overlap years.

The DiD identifying assumption (parallel trends in the absence of treatment) is most plausible for the Indonesia, Nigeria, and Korea events and least plausible for the Brazil, India, and Rwanda events given the documented macro shocks and reform-policy endogeneity in those

windows.

10. Conclusion

Corruption inhibits growth. In our fifteen-country panel from 1998 to 2023, the cross-country relationship between perceived corruption and the country risk premium is statistically significant, with a one-unit deterioration on the inverted WGI scale associated with a 53 percent higher log spread. The implied attribution share is approximately twenty percent of the Africa-Asia spread gap. Total factor productivity accounts for 55.6 percent of the Africa-Asia growth gap, and TFP loads on the inverted WGI with a coefficient of -0.137 ($p = 0.010$), surviving country and year fixed effects.

The institutional-architecture argument finds support in the difference-in-differences analysis of six anti-corruption reform events. Indonesia's KPK and South Korea's consolidation produced statistically significant compressions of the lending-rate proxy. Nigeria's EFCC, instituted in the same year as the KPK and with the same nominal form, produced no measurable response, consistent with documented differences in prosecutorial independence and political-coalition support.

The honest reading is that institutional quality is one of the central drivers of the Africa-Asia growth gap, not the dominant driver. The corruption channel is meaningful and measurable but not principal. The case for institutional reform is strong, the magnitude of the available gains is substantial, and the political-feasibility question (Section 8) is the one that determines whether the gains can be realised in practice.

Future work should incorporate proper EMBI sovereign-spread data, extend the sample to the broader emerging-market panel, and develop the institutional-architecture indicators (prosecutorial independence, judicial follow-through, political-coalition support) into formal cross-country measures. The Open Government Partnership commitment database provides a natural starting point.

References

- Acemoglu, D., Johnson, S., & Robinson, J. A. (2001). The colonial origins of comparative development: An empirical investigation. *American Economic Review*, 91(5), 1369-1401.
- Acemoglu, D., Johnson, S., & Robinson, J. A. (2002). Reversal of fortune: Geography and institutions in the making of the modern world income distribution. *Quarterly Journal of Economics*, 117(4), 1231-1294.
- Acemoglu, D., Johnson, S., & Robinson, J. A. (2003). An African success story: Botswana. In D. Rodrik (Ed.), *In Search of Prosperity*. Princeton University Press.
- Adebawu, W., & Obadare, E. (2011). When corruption fights back: Democracy and elite interest in Nigeria's anti-corruption war. *Journal of Modern African Studies*, 49(2), 185-213.
- Andrews, M. (2013). *The Limits of Institutional Reform in Development*. Cambridge University Press.
- Bandeira-de-Mello, R., Cuervo-Cazurra, A., & Vassolo, R. (2018). Corruption and emerging markets: Evidence from the Brazilian Lava Jato investigation. *Journal of International Business Studies*.
- Bardhan, P. (1997). Corruption and development: A review of issues. *Journal of Economic Literature*, 35(3), 1320-1346.
- Booth, D., & Golooba-Mutebi, F. (2012). Developmental patrimonialism? The case of Rwanda. *African Affairs*, 111(444), 379-403.
- Butt, S. (2009). Anti-corruption reform in Indonesia: An obituary? *Bulletin of Indonesian Economic Studies*, 45(3), 381-394.
- Cameron, A. C., & Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2), 317-372.
- Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200-230.
- Cantor, R., & Packer, F. (1996). Determinants and impact of sovereign credit ratings. *Federal Reserve Bank of New York Economic Policy Review*, 2(2), 37-54.
- Caselli, F. (2005). Accounting for cross-country income differences. In P. Aghion & S. Durlauf (Eds.), *Handbook of Economic Growth*, Vol. 1A. Elsevier.
- de Chaisemartin, C., & D'Haultfoeulle, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9), 2964-2996.
- Easterly, W., & Levine, R. (1997). Africa's growth tragedy: Policies and ethnic divisions. *Quarterly Journal of Economics*, 112(4), 1203-1250.
- Gollin, D. (2002). Getting income shares right. *Journal of Political Economy*, 110(2), 458-474.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2), 254-277.
- Gyimah-Brempong, K. (2002). Corruption, economic growth, and income inequality in Africa. *Economics of Governance*, 3, 183-209.
- Hall, R. E., & Jones, C. I. (1999). Why do some countries produce so much more output per worker than others? *Quarterly Journal of Economics*, 114(1), 83-116.
- Hilscher, J., & Nosbusch, Y. (2010). Determinants of sovereign risk: Macroeconomic fundamentals and the pricing of sovereign debt. *Review of Finance*, 14, 235-262.
- Humphreys, M., & Bates, R. H. (2005). Political institutions and economic policies: Lessons from Africa. *British Journal of Political Science*, 35(3), 403-428.

- Kaufmann, D., Kraay, A., & Mastruzzi, M. (2007). Governance matters VI: Aggregate and individual governance indicators 1996-2006. World Bank Policy Research Working Paper No. 4280.
- Khan, M. (2018). Political settlements and the analysis of institutions. *African Affairs*, 117(469), 636-655.
- Kim, P. S. (2008). A daunting task in Asia: The move for transparency and accountability in the Korean public sector. *Asian Journal of Political Science*, 16(3), 261-280.
- Lambsdorff, J. G. (2003). How corruption affects persistent capital flows. *Economics of Governance*, 4, 229-243.
- La Porta, R., Lopez-de-Silanes, F., & Shleifer, A. (2008). The economic consequences of legal origins. *Journal of Economic Literature*, 46(2), 285-332.
- Levy, B. (2014). *Working with the Grain: Integrating Governance and Growth in Development Strategies*. Oxford University Press.
- Mauro, P. (1995). Corruption and growth. *Quarterly Journal of Economics*, 110(3), 681-712.
- Mauro, P. (2004). The persistence of corruption and slow economic growth. *IMF Staff Papers*, 51(1), 1-18.
- Mota, B., & Sengstacken, A. (2020). Sovereign spread dynamics during the Brazilian political crisis. *Journal of Emerging Market Finance*.
- Olken, B. A. (2007). Monitoring corruption: Evidence from a field experiment in Indonesia. *Journal of Political Economy*, 115(2), 200-249.
- Page, M. T. (2018). *A new taxonomy for corruption in Nigeria*. Carnegie Endowment for International Peace.
- Rodrik, D., Subramanian, A., & Trebbi, F. (2002). Institutions rule: The primacy of institutions over geography and integration in economic development. NBER Working Paper No. 9305.
- Sachs, J. D. (2003). Institutions don't rule: Direct effects of geography on per capita income. NBER Working Paper No. 9490.
- Sachs, J. D., & Warner, A. M. (1995). Natural resource abundance and economic growth. NBER Working Paper No. 5398.
- Schutte, S. A. (2012). Against the odds: Anti-corruption reform in Indonesia. *Public Administration and Development*, 32(1), 38-48.
- Soyombo, F. (2022). The Nigerian anti-corruption record under Buhari, 2015-2022. Premium Times Centre for Investigative Journalism.
- Subramanian, A., & Roy, D. (2003). Who can explain the Mauritian miracle: Meade, Romer, Sachs, or Rodrik? In D. Rodrik (Ed.), *In Search of Prosperity*. Princeton University Press.
- Tanzi, V., & Davoodi, H. (1997). Corruption, public investment, and growth. IMF Working Paper WP/97/139.
- Treisman, D. (2000). The causes of corruption: A cross-national study. *Journal of Public Economics*, 76(3), 399-457.
- Uslaner, E. M. (2008). *Corruption, Inequality, and the Rule of Law*. Cambridge University Press.
- Wedeman, A. (2012). *Double Paradox: Rapid Growth and Rising Corruption in China*. Cornell University Press.
- You, J. (2015). *Democracy, Inequality and Corruption: Korea, Taiwan and the Philippines Compared*. Cambridge University Press.

Appendix

Appendix tables for Sections 6 and 8 of the main text. The full underlying datasets and the analysis code are available in the replication package at github.com/WatchTree-19/africa-institutional-risk-premium.

Appendix Table A1: Anti-corruption reform follow-through dataset (full)

Event ID	Country	Reform yr	Speed	Depth	Survival	Coalition	Composite
IDN_KPK_2003	IDN	2003	3	3	2	2	10
KOR_Consolidation_1995	KOR	1995	3	3	2	2	10
CHN_CCDI_2012	CHN	2012	3	3	2	2	10
VNM_AntiCorr_2016	VNM	2016	3	3	2	2	10
RWA_Ombudsman_2003	RWA	2003	3	2	2	2	9
HKG_ICAC_1991	HKG	1991	3	2	2	2	9
PER_LavaJato_2017	PER	2017	3	2	2	2	9
PHL_Ombudsman_2011	PHL	2011	2	2	2	2	8
BRA_LavaJato_2014	BRA	2014	3	3	1	0	7
ITA_ManiPulite_1992	ITA	1992	3	3	1	0	7
ROU_DNA_2002	ROU	2002	3	3	1	0	7
GTM_CICIG_2007	GTM	2007	3	3	0	0	6
THA_NACC_1997	THA	1997	3	2	1	0	6
ZMB_ACC_2010	ZMB	2010	3	1	1	0	5
SEN_OFNAC_2012	SEN	2012	2	1	1	0	4
NGA_EFCC_2003	NGA	2003	2	1	1	0	4
ARG_AntiCorr_2000	ARG	2000	2	1	1	0	4
IND_Lokpal_2014	IND	2014	0	0	1	0	1
MYS_MACC_2009	MYS	2009	0	0	1	0	1
GHA_OSP_2018	GHA	2018	0	0	1	0	1
MEX_FGR_2019	MEX	2019	0	0	1	0	1
ZAF_StateCapture_2018	ZAF	2018	0	0	1	0	1

Appendix Table A1. Full twenty-two-event follow-through dataset. Speed scored 0-3 (time to first sustained conviction of a senior official; higher = faster). Depth scored 0-3 (count of sustained convictions of senior officials within five years). Survival scored 0-2 (institutional status at year ten: dismantled/weakened/intact). Coalition scored 0-2 (whether the founding political coalition retained effective control through year five). Composite is the weighted sum with weights 3, 3, 2, 2 on the four factors. Source-tier documentation and per-event notes are in the codebook (data/raw/follow_through_codebook.md in the replication package).

Appendix Table A2: Conditional-probability rubric scoring (full)

Country	Electoral	Civil	External	Failure	Exec.	Weighted	Rubric	Case-study
	turnover	society	conditionality	history	constraints	score	band	estimate
Nigeria	1	2	1	0	1	105/200	42-62%	15-25%
Ghana	2	2	2	1	2	185/200	74-94%	30-40%
Senegal	2	2	2	1	2	185/200	74-94%	40-55%
Zambia	2	1	2	0	1	125/200	50-70%	30-40%
Zimbabwe	0	0	0	0	0	0/200	5-20%	5-15%
Rwanda	0	0	1	2	0	45/200	18-38%	framework N/A
Botswana	2	1	1	2	2	165/200	66-86%	60-75%
Mauritius	2	1	2	1	2	165/200	66-86%	45-60%

Appendix Table A2. Full conditional-probability rubric scoring for the eight African sample countries. Factors scored 0 (low), 1 (medium), 2 (high). Weights: 25, 20, 15, 15, 25 percentage points respectively. Weighted score divided by 200 (the maximum) and rescaled to a 10-90 percent range produces the rubric band (with plus-minus ten point uncertainty). Case-study estimates are the country-by-country qualitative judgments from Section 8. Differences between rubric and case-study reflect country-specific structural failure modes the rubric does not index (notably Nigeria's cross-party EFCC weaponisation pattern, Ghana's intermittent coalition reform commitment, the dependence of Senegal's reform trajectory on the 2024 electoral outcome). The rubric is a transparency device; the case-study estimates are the operative figures.